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Solutions development

Learning objective:

17. Understand solutions development as the ability to determine the optimal solution to satisfy agreed requirements.



17a) Understand how to evaluate and prioritise requirements in order to deliver the optimal solution and understand the different approaches for different life cycle models, e.g. the use of MVP and MMP in iterative life cycles.

Evaluating and prioritising requirements

- The business case should document the options considered, including an option to 'do nothing'.
- Solutions development requires clarity about the problem to be solved and then exploring different options until a preferred solution is identified and subsequently maintained and refined.

Things to consider:

- Priority of objectives and requirements.
- Acceptance criteria necessary to be achieved.
- Priorities of the project end users.
- The life cycle approach for project delivery.
- Value of alternative solutions.
- Potential for stakeholder pressure to adopt a 'preferred solution' which may not be the optimal choice.



*The Russian
Space Pen!*

Defining the options

Depending on the type of project, there can be many criteria to think through when defining different options.

- **Technical:** process, material, software, future.
- **Social:** people, interactions, needs, relationships.
- **Procurement:** lease, buy, make, service, partnership.
- **Management:** life cycle, governance.
- **Transition:** phased, all at once 'big bang'.



Photo credit: Glenn Carstens-Peters

Prioritising

- Requirements may need to be prioritised if the project is unable to deliver them all.
- A common technique for this is the MoSCoW approach:
 - **M:** Must have.
 - **S:** Should have.
 - **C:** Could have.
 - **W:** Won't have.
- This looks at how important a requirement is compared to the others.
- The main benefit of the process is that essential requirements are understood, and can then form an input to the work, enabling the selection of the optimal solution.

Solutions development in different life cycle models

Projects using different life cycles take different approaches to developing solutions.

- In a linear life cycle, analysis of options is done early on for the selection of a preferred optimum solution to be agreed with scope. This is then confirmed, and any changes thereafter are managed through change control.
- For an iterative life cycle, there is the opportunity to explore options for longer, allowing different design options to be investigated while the needs are being investigated in another timebox.
- Iterative development provides an environment whereby the solution can be delivered in increments providing opportunity for 'Must have' requirements to be delivered as early as possible, in the form of a minimum viable product (MVP).
- The project can then gain progressive user feedback and the gradual delivery of requirements in the next steps of product development to achieve a minimum marketable product (MMP) through to future versions of the product eventually delivering all requirements.